

# Day 1 – Environment & Project Setup

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## Setup & Prerequisites

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### 1. Setup Checklist:

- **Install Python 3.12+:** Ensure Python 3.12 or higher is installed on your local machine. You can check your version by running `python3 --version` or `python --version` in your terminal.
- **Prepare Your Workspace:** Create an empty project directory on your computer and open a clean terminal window pointing to this folder.
- **Web Browser Ready:** Have a modern web browser open and ready to navigate to the local server address: `http://127.0.0.1:8000/`.

### 2. Prerequisites:

- You must have a working terminal environment suitable for your operating system:
  - **macOS/Linux:** Standard Terminal or Zsh.
  - **Windows:** Git Bash or PowerShell (avoid using command prompt `cmd` as some unix-like commands will not be supported).

## Learning Objectives

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By the end of this class, you will be able to:

- Install and configure modern Python package management using `uv`.
- Initialize and activate a virtual environment.
- Scaffold a new Django project and custom app.
- Securely configure environment variables using `python-decouple`.
- Initialize a Git repository and prevent secrets leaks using `.gitignore`.
- Run the Django development server and verify the default setup.

## Classroom Timeline (45 min)

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| Segment               | Duration | Focus                                                                                |
|-----------------------|----------|--------------------------------------------------------------------------------------|
| 1. Hook & Big Picture | 7 min    | Interactive overview of the bootcamp outcome (game key expiry & publisher webhooks). |

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|------------------------------------------|--------|-----------------------------------------------------------------------------------------|
| <b>2. Key Concepts &amp; Core Ideas</b>  | 7 min  | Introduction to MTV, API backends, package management (uv), and configuration security. |
| <b>3. Live Coding Walkthrough</b>        | 23 min | Live setup of python environment, Django, and decoupling configuration.                 |
| <b>4. Check for Understanding</b>        | 5 min  | Q&A on virtual environments, secrets management, and DEBUG mode.                        |
| <b>5. Class Wrap-up &amp; Checkpoint</b> | 3 min  | Verification of students' local environments.                                           |



## Lecture Step-by-Step Delivery Plan

### 1. Hook & Big Picture (7 min)

The business problem: digital game key marketplaces need to deliver keys to users, track their expirations, and automatically notify publishers when a key expires so they can deactivate it on their servers.

- **Big Picture:** The finished flow: User buys key → key expires → Celery fires webhook → publisher receives it.
- **Q&A:**
  - **Question:** Why do we need webhooks instead of polling?
    - **Answer:** Polling is inefficient because clients have to repeatedly query the database (creating high traffic and latency). Webhooks are event-driven: our system pushes notifications only when an event happens, reducing server load and ensuring near-instant updates.
  - **Question:** Why is this a production-grade problem?
    - **Answer:** It involves cross-system communication, payload security (signing), concurrency (race conditions), reliability (retries, queues), and audit logging.

### 2. Key Concepts & Core Ideas (7 min)

Introduce the following definitions to you:

- **What is Django?** MTV (Model-Template-View) framework. Mention that we'll use it purely as a REST API backend with no HTML templates.
- **What is DRF?** Django REST Framework. A toolkit built on top of Django that gives us serializers, viewsets, and routers for building RESTful APIs.
- **Virtual environments:** Why isolation matters (preventing dependency conflicts and ensuring reproducibility across developers' machines).
  - *Note - Why do we isolate?* Without virtual environments, installing packages globally can overwrite

dependencies needed by other projects on the same machine, causing conflicts and breaking applications.

- **DEBUG=True vs False:** Django displays highly detailed error pages containing traceback information. Explain that this is crucial for development but a critical security leak in production.
  - *Note - Why is it a leak?* It exposes database queries, settings, system file paths, configurations, and environment variables to the end user.
- **Why python-decouple?** Reading from `os.environ.get()` is functional but `decouple` simplifies casting types (e.g. converting a string `"True"` to boolean `True`) and handling defaults and local `.env` files in a clean API.

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### 3. Live Coding Walkthrough (23 min)

#### Step 3.1: Tooling Installation & Scaffolding

- **Concept:** Contrast standard `pip + venv` with `uv`. We use `uv` for speed, lockfiles, and reproducibility. `uv` is an extremely fast Python package installer and resolver written in Rust. It serves as a drop-in replacement for `pip`, `pip-tools`, and `virtualenv`. Run the installation and scaffolding commands.

- **Code:**

```
# Install uv (macOS/Linux)
# This command downloads the uv binary installation script and executes it via the shell.
curl -LsSf https://astral.sh/uv/install.sh | sh

# Create and activate virtual environment
# uv venv creates a local virtual environment named '.venv' in the current working directory.
uv venv

# Activate the environment so your terminal uses the virtual environment's Python interpreter
source .venv/bin/activate

# Add dependencies
# uv add automatically adds packages to your environment and manages a project lockfile.
# Packages installed:
# - django: The core web framework.
# - djangorestframework: Toolkit for building RESTful APIs on top of Django.
# - python-decouple: Utility for separating settings configuration from source code.
# - celery: Task runner for asynchronous and background execution.
# - redis: Python client for connecting to the Redis database (used as Celery's broker).
# - pytest-django: Django integration helper for testing with pytest.
uv add django djangorestframework python-decouple celery redis pytest-django

# Scaffold project and app
# django-admin startproject creates a folder structure containing core settings.
# The '.' at the end is CRITICAL: it creates the project in the CURRENT directory, preventing
django-admin startproject gamekey_platform .

# startapp creates a custom Django application named 'games' containing models, views, and con
```

```
python manage.py startapp games
```

- **Verify:** Confirm that you see `(.venv)` in your command line prompt. Ensure you are inside the active virtual environment before moving on. Run `which python` (or `where python` in Windows PowerShell) to verify it points to the `.venv` directory. ⚠ **Common Pitfalls I will flag:**
- **Forgetting to activate the venv:** If you skip `source .venv/bin/activate`, you will install dependencies globally or get "django not found". Run `which python` to verify the path points to `.venv`.
- **Windows path differences:** Note that on Windows, the activation command is `.venv\Scripts\activate` rather than `source .venv/bin/activate`.

### Step 3.2: Environment Variable Configuration

- **Concept:** Secrets (such as the Django `SECRET_KEY` or database passwords) should never go into version control (source code). If you push a secret key to a public GitHub repository, bots will scan it within seconds and exploit it. We extract these values into a local environment file called `.env`.
- **Code:** Create `.env` in the project root directory:

```
SECRET_KEY=your-secret-key-here
DEBUG=True
```

- **Verify:** Check that the `.env` file is created and contains the correct key-value pairs.

### Step 3.3: Django settings.py Decoupling

- **Concept:** Now we need to modify Django's settings to read our newly created `.env` file rather than having hardcoded variables. `python-decouple` provides a helper function `config()` that searches for the variable in the `.env` file first, and falls back to system environment variables if not found.
- **Code:** Update `gamekey_platform/settings.py` to import and use `python-decouple`:

```
# Add this import at the top of the file
from decouple import config

# Replace the hardcoded SECRET_KEY line with this:
SECRET_KEY = config('SECRET_KEY')

# Replace the hardcoded DEBUG line with this:
# The default is set to False for safety, and cast=bool converts the environment string "True"
DEBUG = config('DEBUG', default=False, cast=bool)
```

- **Verify:** Run the development server to verify the configuration is loaded correctly:

```
# Verify setup
python manage.py runserver
```

Navigate to `http://localhost:8000/` and verify the Django welcome page loads. ⚠ **Common Pitfalls I will flag:**

- **SECRET\_KEY still hardcoded:** You might add `python-decouple` but forget to update the actual variables in `settings.py`. Test this by removing the key from `.env` and seeing if Django throws an error on startup.

### Step 3.4: Git Initialization

- **Concept:** Set up version control. Introduce the `.gitignore` file and emphasize that the `.env` file must be ignored immediately to prevent leaking the secret keys.

- **Code:**

```
git init
```

Create `.gitignore` and include `.env` (along with standard Python/Django patterns: `__pycache__`, `.venv`, `*.pyc`, etc.):

```
.env
.venv/
*.pyc
__pycache__/
db.sqlite3
```

- **Verify:** Run `git status` and confirm that `.env` is NOT listed under untracked files. ⚠ **Common Pitfalls I will flag:**
- **Committing .env to Git:** Check what `git status` looks like *before* and *after* editing `.gitignore` to see how Git tracks files.

## 4. Check for Understanding (5 min)

**Question:** "Why can't we just hard-code `SECRET_KEY = 'abc123'` in `settings.py`?"

**Answer:** Hard-coding `SECRET_KEY` in source code poses a severe security risk. If the repository is pushed to a public platform (like GitHub), anyone can see the key. Attackers can use it to forge signed cookies, session data, or password reset tokens. Storing it in an environment variable (`.env` file) keeps it private and allows different values for development and production environments.

**Question:** "What would happen if two projects on the same machine installed different versions of `django` without virtual environments?"

**Answer:** Installing different versions globally would cause a conflict because Python can only resolve one version of a package in its global `site-packages` directory. Installing a different version for the second project would overwrite the version needed by the first project, breaking it. Virtual environments isolate dependencies per project, allowing each to run its required version

independently.

**Question:** "What does `DEBUG=True` change about how Django behaves?"

**Answer:** When `DEBUG=True`, Django displays detailed error traceback pages to the client (useful for debugging but a huge security leak in production because it exposes database queries, settings, and path variables). It also runs the built-in development server's auto-reloader and serves static/media files automatically. In production (`DEBUG=False`), it returns a generic 500 error page, requires `ALLOWED_HOSTS` to be set, and disables automated static file serving.

## 5. Daily Checkpoint & Wrap-up (3 min)

- **Verification Criteria:**

- ☐ You can access the Django welcome rocket page running locally at `http://127.0.0.1:8000/`.
- ☐ The `.env` file exists and contains `SECRET_KEY` and `DEBUG`.
- ☐ Running `git status` shows `.env` is untracked (not staged) and excluded.
- ☐ Running `git log --oneline` shows at least one commit.